

11. ResultSetMetaData interface

The ResultSetMetaData interface in Java is part of the java.sql package and provides methods to retrieve metadata about the columns of a ResultSet. This is useful when you need to dynamically understand the structure of query results, especially for generic database operations.

The metadata means data about data i.e. we can get further information from the data. If you have to get metadata of a table like total number of column, column name, column type etc. , ResultSetMetaData interface is useful because it provides methods to get metadata from the ResultSet object.

Key Features of ResultSetMetaData

- Provides column names, types, and sizes.
- Indicates whether columns can be null or are auto-incremented.
- Useful for building generic database tools or utilities.

Commonly Used Methods of ResultSetMetaData Interface

Method	Description
int getColumnCount()	Returns the number of columns in the result set.
String getColumnName(int column)	Returns the name of the specified column (1-based index).
String getColumnType(int column)	Returns the database-specific type name of the specified column.
int getColumnDisplaySize(int column)	Returns the suggested column width for displaying the column value.
int isNullable(int column)	Indicates whether the column allows NULL values (columnNullable, columnNoNulls, or columnNullableUnknown).
boolean isAutoIncrement(int column)	Indicates whether the column value is automatically incremented.
boolean isReadOnly(int column)	Indicates if the column is read-only.
boolean isWritable(int column)	Indicates if the column can be written to.

Example 1: Fetching Metadata for a Table

```
package com.lokesh;

import java.sql.*;

public class ResultSetMetaDataInfo {
    public static void main(String[] args) {
        String jdbcUrl = "jdbc:mysql://localhost:3306/organization";
        String username = "root";
        String password = "root";
```

```

String query = "SELECT * FROM DateTimeExample";

Connection connection = null;
Statement statement = null;
ResultSet resultSet = null;

try {
    // Establish connection
    connection = DriverManager.getConnection(jdbcUrl, username, password);
    statement = connection.createStatement();
    resultSet = statement.executeQuery(query);

    // Get metadata
    ResultSetMetaData metaData = resultSet.getMetaData();
    int columnCount = metaData.getColumnCount();

    System.out.println("Column Metadata:");
    for (int i = 1; i <= columnCount; i++) {
        System.out.println("Column " + i + ":");
        System.out.println("  Name: " + metaData.getColumnName(i));
        System.out.println("  Type: " + metaData.getColumnTypeName(i));
        System.out.println("  Display Size: " + metaData.getColumnDisplaySize(i));
        System.out.println("  Nullable: " + (metaData.isNullable(i) ==
            ResultSetMetaData.columnNullable ? "YES" : "NO"));
        System.out.println("  Auto Increment: " + metaData.isAutoIncrement(i));
        System.out.println("-----");
    }

} catch (SQLException e) {
    e.printStackTrace();
} finally {
    // Close resources explicitly
    try {
        if (resultSet != null) resultSet.close();
        if (statement != null) statement.close();
        if (connection != null) connection.close();
    } catch (SQLException e) {
        e.printStackTrace();
    }
}
}

```

Output:

```

Column Metadata:
Column 1:
  Name: id
  Type: INT
  Display Size: 10
  Nullable: NO
  Auto Increment: true
-----
Column 2:
  Name: event_date
  Type: DATE
  Display Size: 10
  Nullable: NO
  Auto Increment: false
-----
Column 3:
  Name: event_time
  Type: TIME
  Display Size: 10
  Nullable: NO

```

```

Auto Increment: false
-----
Column 4:
  Name: event_timestamp
  Type: TIMESTAMP
  Display Size: 19
  Nullable: YES
  Auto Increment: false
-----

```

Example 2: Dynamic Data Printing

```

package com.lokesh;
import java.sql.*;

public class DynamicTablePrinter {
    public static void main(String[] args) {
        String jdbcUrl = "jdbc:mysql://localhost:3306/organization";
        String username = "root";
        String password = "root";

        String query = "SELECT * FROM DateTimeExample";

        Connection connection = null;
        Statement statement = null;
        ResultSet resultSet = null;

        try {
            // Establish connection
            connection = DriverManager.getConnection(jdbcUrl, username, password);
            statement = connection.createStatement();
            resultSet = statement.executeQuery(query);

            // Get metadata
            ResultSetMetaData metaData = resultSet.getMetaData();
            int columnCount = metaData.getColumnCount();

            // Print column headers
            for (int i = 1; i <= columnCount; i++) {
                System.out.print(metaData.getColumnName(i) + "\t");
            }
            System.out.println();

            // Print rows
            while (resultSet.next()) {
                for (int i = 1; i <= columnCount; i++) {
                    System.out.print(resultSet.getString(i) + "\t");
                }
                System.out.println();
            }
        } catch (SQLException e) {
            e.printStackTrace();
        } finally {
            // Close resources explicitly
            try {
                if (resultSet != null) resultSet.close();
                if (statement != null) statement.close();
                if (connection != null) connection.close();
            } catch (SQLException e) {
                e.printStackTrace();
            }
        }
    }
}

```

```
}  
}
```

Output:

id	event_date	event_time	event_timestamp
1	2024-12-04	14:30:00	2024-12-04 14:30:00